



Optical properties and thermal stability of germanium oxide (GeO₂) nanocrystals with α -quartz structure

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ABSTRACT

Germanium dioxide (GeO₂) crystals were prepared by a chemical precipitation method at a relatively low-temperature (100 °C). The grown crystals were characterized by studying their microstructure, optical properties and thermal stability. The results indicate that the grown GeO₂ crystals exhibit α -quartz type crystal structure. The lattice parameters obtained from XRD were $a = 4.987(4)$ Å and $c = 5.652(5)$ Å. Electron microscopy analysis indicates a high structural quality of GeO₂ crystals grown using the present approach. Optical absorption measurements indicate a direct bandgap of 5.72 eV without any additional bands arising from localized or defect states. Thermogravimetric measurements indicate the temperature stability of the grown GeO₂ nanocrystals. Microscopic analysis coupled with energy dispersive X-ray spectroscopy of the GeO₂ crystals with α -quartz type crystal structure indicates their stability in chemical composition up to a temperature of 400 °C. The surface morphology of GeO₂ crystals, however, found to be changing with the increase in temperature.

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1. Introduction

The growth, characterization, and utilization of metal-oxide crystals for application in modern electronic and optical devices continue to be an interesting topic for theoretical and experimental investigations. Germanium dioxide (GeO₂) exhibits many interesting physiochemical properties for applications in optical, electronic and optoelectronic devices [1–15]. GeO₂ is a photoluminescence and dielectric material. It exhibits high values of dielectric constant, refractive index, thermal stability and mechanical strength. Photo-sensor capability of GeO₂ has been confirmed by optical spectroscopic methods such as Raman-spectroscopy and electron spin resonance spectroscopy coupled with structural measurements [1–3]. Due to these fascinating optical and electronic properties, GeO₂ has been considered as a promising material for optical waveguides and nano-connections in optoelectronic communications.

Synthesis and optimization of a particular phase of GeO₂ crystals is very important as this material exhibits several polymorphs [15–21]. The tetragonal α -phase with rutile type structure

($P4_2/mnm$, PDF 35-729 [15,16]), in which Ge is six-fold coordinated, is stable up to 1047 °C. The trigonal/rhombohedral β -phase with α -quartz type [17–20] ($P3_121$, PDF 04-498 [17–18] and $P3_221$, PDF 36-1463 [19,20]), in which Ge is four-fold coordinated, is stable above 1035 °C. Several high-pressure phases and glassy structures of GeO₂ were also reported in the literature [21]. GeO₂ crystals are usually produced by either physical evaporation or thermal oxidation methods. Crystallization through these techniques leads to the formation of one-dimensional structures [4–7]. Attempts were made to obtain the nanofibers by electrospinning process [8]. The GeO₂ whiskers were fabricated by ablating Ge target at $T = 820$ °C with a pulsed KrF excimer laser in an argon atmosphere [9]. Chemical precipitation and hydrothermal methods were also employed to grow three-dimensional GeO₂ structures with sizes from 15 nm to several centimeters depending on synthesis conditions [10–13]. Wang et al. have studied the ion-irradiation induced polymorphs of GeO₂ [22]. Hexagonal α -phase GeO₂ crystal formation on the Ge single crystal surfaces due to acid vapor exposure was reported by Kalem et al. [23]. Kim et al. demonstrated the growth of uncommon cone-shaped structures of GeO₂ using a thermal evaporation process [14]. In present study β -phase GeO₂ crystals with α -quartz type structure were grown using a low-temperature ($T = 100 \pm 5$ °C) chemical precipitation method. The microstructural and morphological characteristics of the obtained nanocrystals were reported elsewhere [24]. The optical properties and thermal stability of β -

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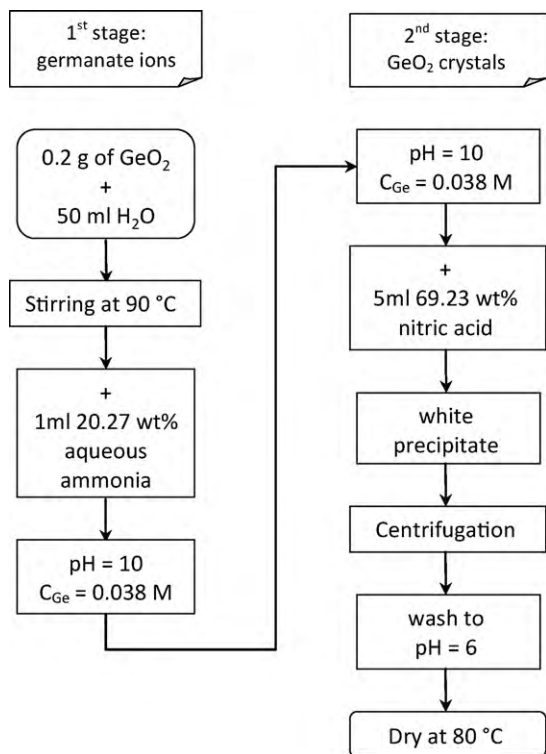


Fig. 1. Schematic flow chart indicating the synthesis procedure of GeO₂ crystals.

phase GeO₂ crystals with α -quartz type structure are reported in this paper.

2. Experimental

The starting materials were GeO₂ powder (Umicore Electro-optic Materials) with optical grade, aqueous ammonia (20.27 wt %), nitric acid (69.23 wt %) and distilled water. The synthesis procedure adopted is schematically represented in a flow chart as shown in Fig. 1. Two steps were carried out to obtain a GeO₂ precipitate. At first 0.2 g of GeO₂ was added to 50 ml distilled water and heated under magnetic stirring to a temperature of 90 °C. Then 1.6 ml aqueous ammonia was slowly added into the solution to form an ammonium germanate aqua solution with a pH 10 and germanium concentration $C_{\text{Ge}} = 0.038$ mol/l. Under the above conditions the reaction between crystalline GeO₂ and aqueous ammonia produces germanate ions as monomers [25]. At second stage, 5 ml nitric acid was added to ammonium germanate solution and white precipitate appeared when the solution was evaporated down to 10 ml volume. The precipitate obtained was separated by centrifugation and washed with distilled water several times to pH 6 of washing water. The final product was dried in air at 80 °C.

The microstructure and morphology of GeO₂ nanocrystals was determined by scanning electron microscopy (SEM) using the Hitachi S4800 high-performance machine. The secondary electron imaging was employed to obtain the images. The energy dispersive X-ray spectrometry (EDS) analysis was also performed in the same instrument. The X-rays generated in the sample by the electron beam bombardment of the sample were employed to obtain the chemical information. Structural properties of germanium oxide were evaluated with X-ray diffraction (XRD) analysis. XRD patterns were recorded by DRON-3M device with Cu K α radiation, Ni filter, counting 10 s per step, step size 0.02°. A Shimadzu UV-3101 PC UV-vis-NIR spectrometer was used to collect the absorption spectra of GeO₂ crystals. The measurements were performed on samples loaded into a quartz cuvette. Thermogravimetric and dif-

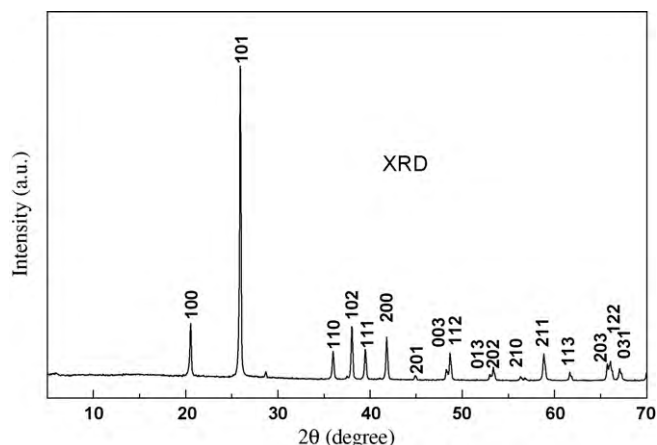


Fig. 2. XRD pattern of GeO₂ crystals.

ferential scanning calorimetric analysis (TG/DSC) was performed with Netzsch STA 449C device. The measurements were carried out using a heating rate of 10 °C/min between 20 and 1000 °C in air.

3. Results and discussion

The powder X-ray diffraction pattern of the GeO₂ sample obtained is shown in Fig. 2. XRD curve reveals good crystallinity and phase purity of as-prepared β -GeO₂. All the diffraction peaks were successfully indexed and their positions are in good agreement with those of the β -GeO₂ with α -quartz type structure (space group $P3_212$) with lattice constants $a = 4.985$ Å and $c = 5.648$ Å. The detailed analysis and comparison of the XRD and TEM data along with a comparison of the d -spacing values is reported elsewhere [24]. Phase composition of the precipitate as β -phase GeO₂ with α -quartz type ($P3_212$, PDF 43-1016) has been determined by both XRD and TEM analyses where the presence of only β -phase of GeO₂ was evident [24].

The SEM and EDS results are shown in Fig. 3. SEM images indicate the formation of white hexagonal, rounded or faced monosized GeO₂ nanocrystals with typical diameter up to 500 nm. The EDS spectrum obtained from the sample exhibits the characteristic peaks corresponding to Ge and O atoms. The X-rays emitted from the sample as a result of electron beam bombardment are characteristic of individual atoms [26–28]. The EDS measurements, therefore, can be used to qualitatively discuss the chemical characteristics of the grown samples. The EDS curve displays the peaks due to Ge K, Ge L and O K lines. The absence of any other peaks except those due to Ge and O is evidence that the sample consists exclusively of Ge oxide phase with no other detectable impurities from other elements. The XRD, SEM and EDS results indicate that the GeO₂ crystals synthesized exhibit high structural quality in terms of physical and chemical structure.

The optical properties of GeO₂ nanocrystals measured in the UV-vis range were used to further access their quality and electronic structure information. The optical absorption curve of GeO₂ crystals is shown in Fig. 4. The absorption measurements in the UV-visible range indicate that the GeO₂ samples show essentially no absorption in the spectral region above 300 nm. The onset of a very strong optical absorption at ~ 220 nm, as seen in the curve (Fig. 4), is due to the fundamental absorption within the bandgap of the material. The absence of any other bands or sub-bands or band splitting in the optical absorption curve must be noted for the GeO₂ crystals with no impurities or vacancies.

The optical bandgap of GeO₂ is evaluated using the optical density (OD) vs. energy (E) plot. The optical density is calculated using

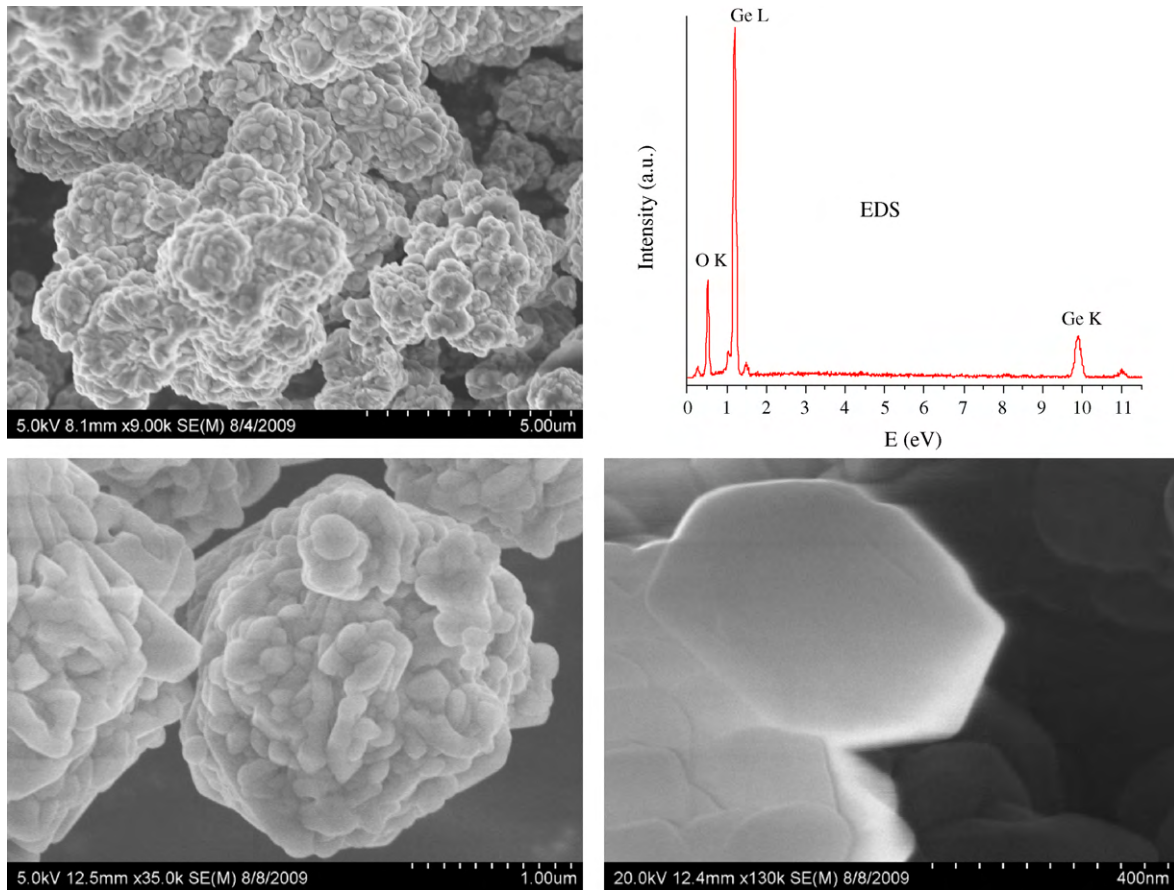


Fig. 3. SEM and EDS data of as-grown GeO_2 crystals.

the relation [29]:

$$\text{OD} = \log \left[\frac{1}{1-A} \right] \quad (1)$$

where 'A' is absorbance. A better fit of the data is obtained when $(\text{OD})^2$ is plotted against the incident photon energy (E) as shown in Fig. 5. This suggests the direct bandgap nature of the GeO_2 crystals

grown in this work. The bandgap is evaluated by extrapolating the linear region of the plot to zero-absorption (Fig. 5). The evaluated optical bandgap is 5.72 eV. The optical properties of α -quartz type structured GeO_2 are rarely mentioned in the literature. Therefore, a direct comparison of our results is not readily possible. However, the two important points that can be derived from our data are as follows. The first is the direct bandgap of the material, which is in good agreement with the other report in the literature [30]. The

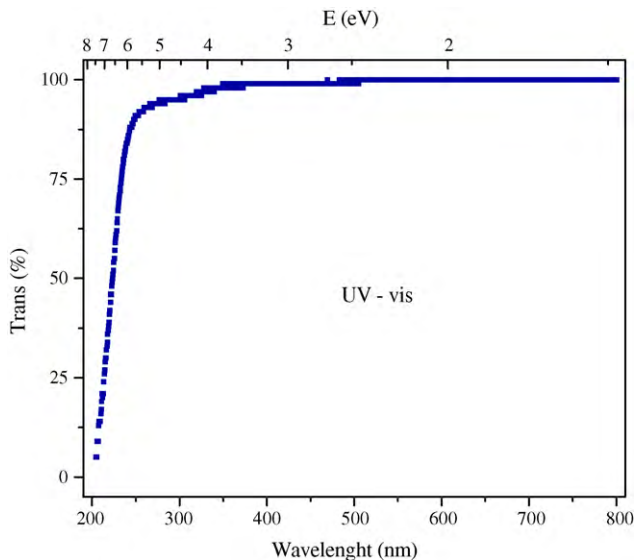


Fig. 4. Optical absorption spectrum of GeO_2 crystals. The onset of strong absorption at ~ 220 nm is due to fundamental absorption across the bandgap of the material.

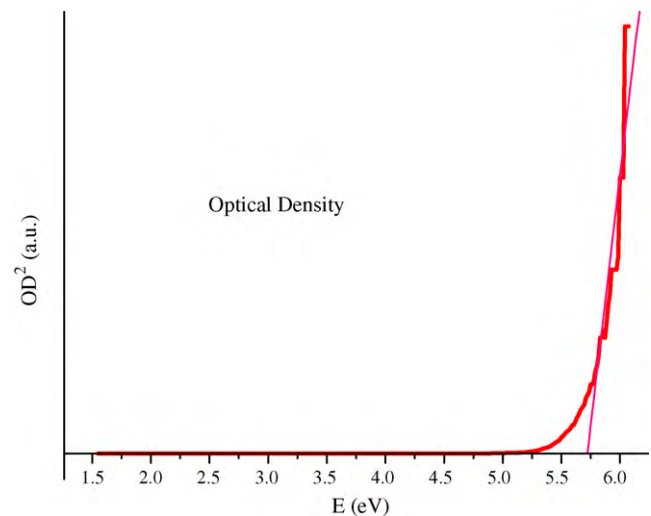


Fig. 5. Plot of $(\text{OD})^{1/2}$ vs. energy for GeO_2 crystals. The better linear fit indicates the direct bandgap of GeO_2 . The bandgap is determined by extrapolating the linear region.

second is the magnitude (5.72 eV in this work) and its correlation to the structure. Lange et al. investigated the optical properties of GeO_x films and found a direct correlation of the optical bandgap to the flow rates of oxygen and, hence, the oxygen content in the films [30]. It was reported that the GeO_x films grown at O_2 flow above 20 sccm and below 25 sccm exhibit the bandgap in the visible region. However, when the GeO_x films are grown O_2 flow more than 25 sccm, the bandgap shifts to >5 eV. This was attributed the oxygen vacancies, which decreases the bandgap significantly. Most importantly, an increase in the bandgap with increasing oxygen flow and, hence, oxygen stoichiometry in GeO_x and a bandgap 5.9 eV for α - GeO_2 (quartz-structure) (in the region of saturation O_2 flow) were reported [30]. Taking these reported results into consideration and comparing with our own optical data, it is clear that the GeO_2 materials grown in the work exhibit a high chemical and optical quality. The optical bandgap of GeO_2 with α -quartz structure (5.7 eV) is in good agreement with the reported value for the same phase.

Thermogravimetric (TG) and differential scanning calorimetry (DSC) curves of GeO_2 are shown in Fig. 6. An examination of the curves indicate that adsorption of water contents is ~ 0.2 wt.% and this component is removed up to a temperature of 180°C . A 2% increase in weight occurs in the temperature range 225 – 275°C accompanied with a small exothermic effect at $T=275^\circ\text{C}$. A 2% increase in weight could be due to the oxygen vacancy filling. Further increase in temperature results in a gradual reduction in weight by $\sim 4\%$ together with two endothermic effects at temperatures 400 and 440°C , respectively, and two small exothermic effects at 600 and 935°C , respectively. Two endothermic effects at temperatures 400 and 440°C can be due to a motion of structural units (tetrahedrons) with loss of OH^- groups whose content

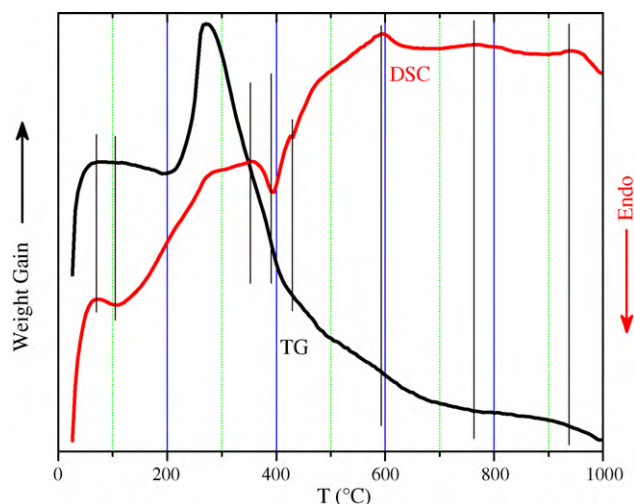


Fig. 6. TG and DSC curves of GeO_2 crystals.

is above 4% and decreases gradually up to 1000°C . This motion of tetrahedrons is also confirmed by changes of crystal surface morphology in discussed in SEM measurements of thermal stability. Two small exothermic peaks at 600 and 935°C may be due to small changes in GeO_4 tetrahedra tilting under temperature increase, for example due to loss of residual OH groups. The α -quartz structure type remains the same within the range of temperature. In order to confirm the stability of the GeO_2 crystals, a combined SEM and EDS analysis was performed as a function of temperature. The results

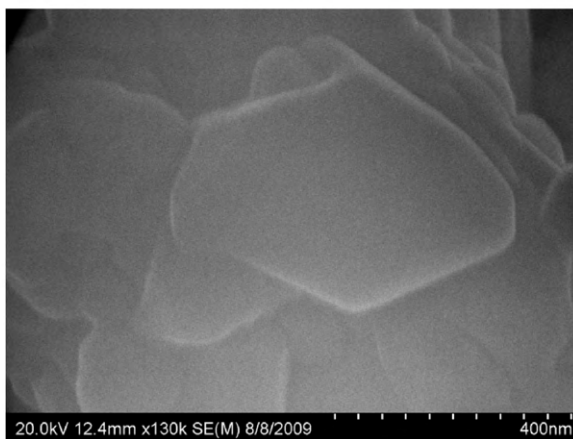
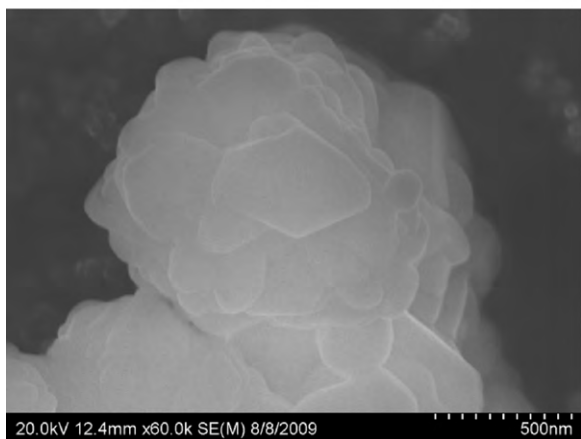
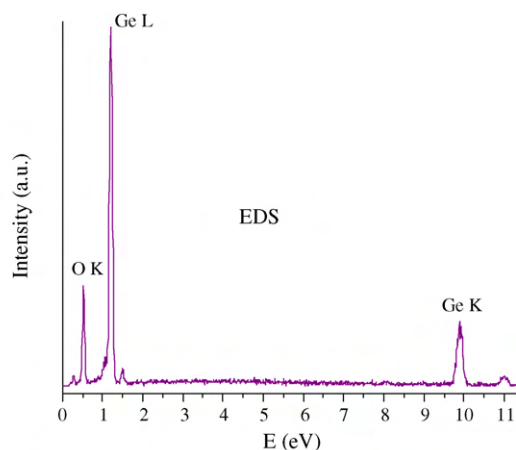
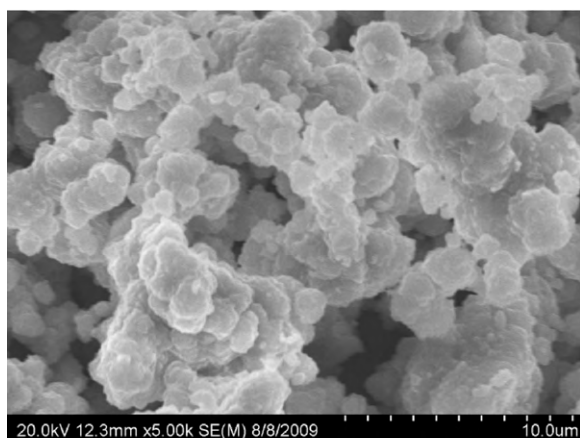


Fig. 7. SEM and EDS results of GeO_2 crystals heated to a temperature of 200°C .

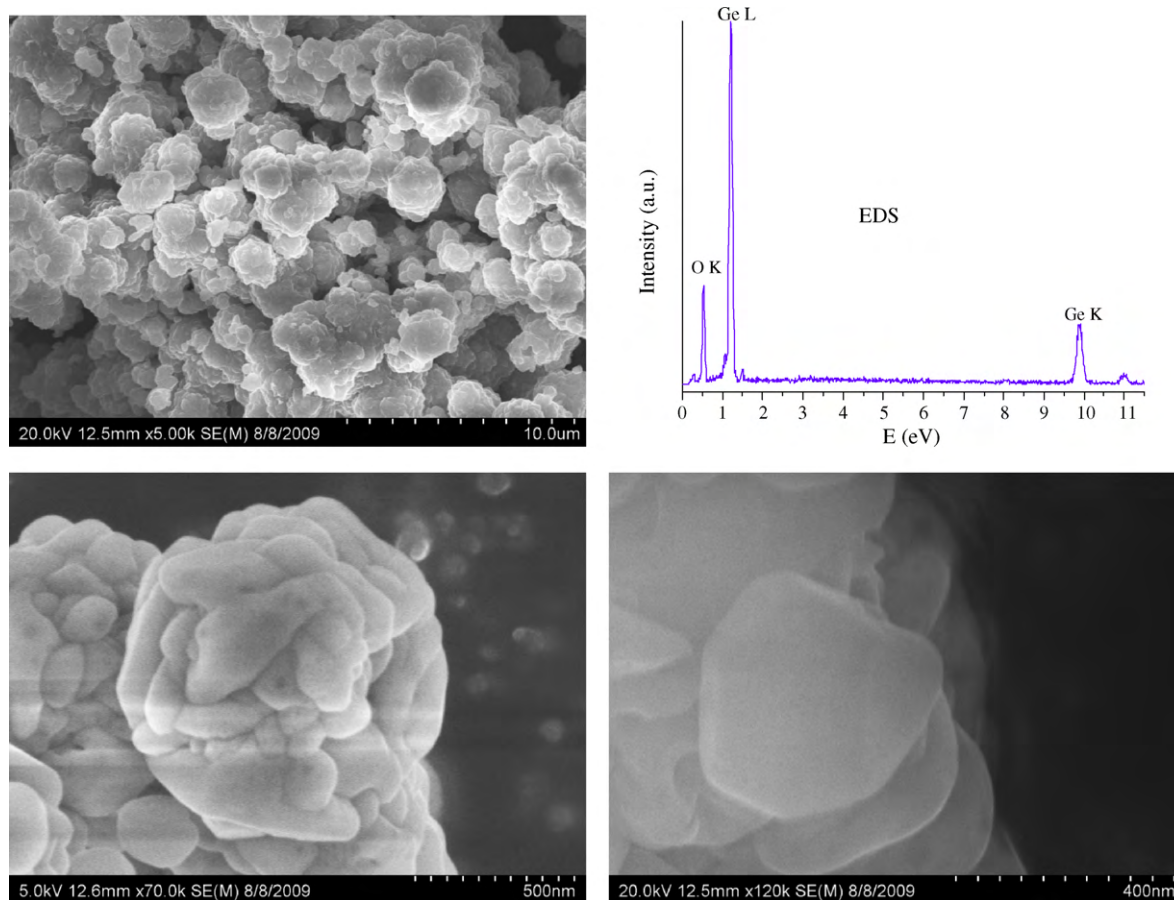


Fig. 8. SEM and EDS results of GeO_2 crystals heated to a temperature of 400°C .

are shown in Figs. 7 and 8. As seen in the EDS curves (Fig. 8), no significant changes were noticed in the chemical composition up to the highest temperature (400°C) employed in this work. However, it is clear that the crystals are changing their morphology with the increase in temperature. GeO_2 samples are agglomerated into a network of hexagonal shaped crystals (Fig. 3). It is important to notice the characteristic feature of agglomeration into hexagonal shape while retaining the same shape of the individual crystals. Faceting and agglomeration of the crystals depends on the specific energies associated with the crystals planes or directions. It seems that GeO_2 crystals in this case have a tendency to form hexagonal shaped individual crystals and also network of crystals into hexagonal shape to arrange their best possible configuration (Fig. 3). At a temperature of 200°C , crystals tend to lose the agglomeration and hexagonal shape of the network (Fig. 7). However, there is no significant change in either size or hexagonal shape of the individual crystals. A temperature of 200°C might be providing the sufficient energy for the crystals to lose integrity. It is very clear that the crystals change their morphology to a greater extent at temperature of 400°C (Fig. 8). The crystals are well separated from the hexagonal network of agglomeration. However, they now tend to agglomerate into a network of rounded and flower petals type of architecture. At the same time, the individual crystals also lose their hexagonal shape and trying to become rounded. The mixture of circular and hexagonal shaped crystals can be seen in the SEM images for samples heated to 400°C (Fig. 8).

4. Conclusions

$\beta\text{-GeO}_2$ nanocrystals, which are traditionally formed at temperatures higher than 1050°C , were synthesized for the first time by

a simplified wet chemical method at relatively low-temperatures, as low as 100°C . Characterization of the crystals using microscopic and spectroscopic methods reveals that the $\beta\text{-GeO}_2$ nanocrystals are single phase with a well-controlled morphology. The crystals are uniform in size and shape with obtained lattice parameters: $a = 4.987(4)\text{\AA}$ and $c = 5.652(5)\text{\AA}$. The SEM and EDS measurements indicate that these crystals are chemically stable up to 400°C , as measured in this study. However, the crystals tend to change their surface morphology, specifically from hexagonal shape at room temperature into a circular shape at 400°C . The sequence of transition in this surface morphology initiates at a temperature of 200°C . The high-temperature stability and the effect of changes in the morphology on the electronic properties will be evaluated in future. The optical absorption measurements indicate a direct bandgap of 5.72 eV for $\beta\text{-GeO}_2$ crystals. The measurements also indicate the highly photoluminescent phase GeO_2 with excellent optical quality. These results suggest that the grown GeO_2 crystals could be useful for optical and optoelectronic applications.

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